

VARDHAMAN COLLEGE OF ENGINEERING

DEPARTMENT OF CSE

Mini Project Review - I

Title: Predictive Analytics for Cybercrime Complaints

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TITLE

- Project Title: Predictive Analytics for Cybercrime Complaints
- Domain/Area: Web Based App & Machine Learning
- Type: Research + Application
- Keywords: Extract features from URL, Train a Random Forest or SVM model on a dataset, Cybercrime prediction, Crime pattern forecasting

BACKGROUND RESEARCH

A browser extension or a web app that analyzes a URL in real-time to determine if it is legitimate or a phishing attempt. Why it's impressive: Cybersecurity is a hot field. Using ML to predict security threats shows you can apply data science to real-world safety.

- Brief overview of existing systems or solutions
 - Most current cybercrime systems focus on **reporting and post-incident investigation**. Tools like cybercrime portals, SIEM platforms, and basic dashboards provide incident logs, alerts, and historical statistics. While useful, they mainly support **reactive cybersecurity**, offering limited forecasting or predictive insights.

- Identify gaps or challenges in existing work
 - **No predictive analysis** to forecast future cybercrime trends or hotspots.
 - **Limited integration** of temporal, spatial, and behavioral data.
 - **Lack of unified frameworks** combining classification, forecasting, and geospatial modeling.
 - **Dashboards are mostly static**, offering minimal actionable intelligence.
- Motivation behind the proposed work
 - Cybercrime is increasing quickly, requiring a move from reactive to proactive approaches. A predictive system can help law enforcement anticipate incidents, identify hotspots, and allocate resources better. This project addresses existing gaps by offering integrated predictions and an interactive dashboard for smarter decision-making.

PROJECT PLANNING

PREDICTIVE ANALYTICS FOR CYBERCRIME COMPLAINTS

GANTT CHART

PROJECT TIMELINE



LITERATURE SURVEY

Paper Title	Approach/Algorithm	Dataset Used	Key Findings	Gaps Identified
Computational System to Classify Cyber Crime Offenses using Machine Learning (2020)	Support Vector Machine (SVM) , Random Forest, and Decision Trees for text classification.	Kaggle Cybercrime Datasets.	SVM achieved the highest accuracy in distinguishing between "Violent" and "Non-Violent" .	The model relies heavily on structured data and struggles with unstructured text without heavy pre-processing.
Cybercrime: Identification and Prediction Using Machine Learning Techniques (2022)	K-Means Clustering (Unsupervised Learning) & Gaussian Mixture Models.	CBS Open Data StatLine (National Crime Statistics).	Unsupervised clustering identified new patterns of fraud that did not fit into pre-existing categories.	The detection rate for unsupervised methods (76%) was significantly lower than supervised methods.

OBJECTIVES

- State clear and measurable objectives:
 1. To develop a predictive model that forecasts future cybercrime incidents using historical complaint data.
 2. To identify geospatial hotspots by applying clustering techniques to detect high-risk regions.
 3. To build an interactive dashboard that visualizes predictions, trends, and risk zones for improved decision-making.

PROJECT MILESTONE TRACKER

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Q&A